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Film scheme to reduce plasma-induced damage

Abstract

The present disclosure relates to an integrated chip structure. The integrated chip structure includes a substrate. One or more lower interconnects are disposed within a lower inter-level dielectric (ILD) structure over the substrate. A plasma induced damage (PID) mitigation layer is disposed over the lower ILD structure. The PID mitigation layer has a porous structure including a metal. A first upper interconnect is laterally surrounded by an upper ILD structure over the PID mitigation layer. The first upper interconnect extends from over the PID mitigation layer to the one or more lower interconnects.

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Background/Summary

REFERENCE TO RELATED APPLICATION (1) This Application claims the benefit of U.S. Provisional Application No. 63/332,900, filed on Apr. 20, 2022, the contents of which are hereby

incorporated by reference in their entirety.

BACKGROUND

(1) Integrated chip fabrication is a complex multiple-step process during which electronic circuits are formed on a wafer made out of a semiconductor material (e.g., silicon). Integrated chip fabrication can be broadly divided into front-end-of-line (FEOL) processing and back-end-of-line (BEOL) processing. FEOL processing generally relates to the formation of devices (e.g., transistors) within the semiconductor material, while BEOL processing generally relates to the formation of conductive interconnects within a dielectric structure over the semiconductor material.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
- (2) FIG. 1 illustrates a cross-sectional view of some embodiments of an integrated chip structure comprising a plasma induced damage (PID) mitigation layer configured to reduce plasma induced damage.
- (3) FIG. 2 illustrates a cross-sectional view of some additional embodiments of an integrated chip structure comprising a PID mitigation layer.
- (4) FIG. 3 illustrates a cross-sectional view of some additional embodiments of an integrated chip structure comprising a plurality of PID mitigation layers.
- (5) FIG. 4 illustrates a cross-sectional view of some additional embodiments of an integrated chip structure comprising a PID mitigation layer.
- (6) FIGS. 5A-5B illustrate cross-sectional views of some additional embodiments of integrated chip structures comprising a PID mitigation layer.
- (7) FIG. 6 illustrates a cross-sectional view of some additional embodiments of an integrated chip structure comprising a PID mitigation layer having a plurality of sub-layers.
- (8) FIG. 7 illustrates a cross-sectional view of some additional embodiments of an integrated chip structure comprising a PID mitigation layer.
- (9) FIG. 8 illustrates a cross-sectional view of some additional embodiments of an integrated chip structure comprising a MIM structure having a PID mitigation layer.
- (10) FIG. 9 illustrates a cross-sectional view of some additional embodiments of an integrated chip structure comprising a plurality of PID mitigation layers.
- (11) FIGS. 10A-10B illustrate graphs showing exemplary gate leakage currents for transistor devices on an integrated chip structure having a disclosed PID mitigation layer.
- (12) FIGS. 11-20 illustrate some embodiments of a method of forming an integrated chip structure comprising a PID mitigation layer.
- (13) FIGS. 21-28 illustrate some additional embodiments of a method of forming an integrated chip structure comprising a PID mitigation layer.
- (14) FIGS. 29-34 illustrate some additional embodiments of a method of forming an integrated chip structure comprising a PID mitigation layer.
- (15) FIGS. 35-42 illustrate some additional embodiments of a method of forming an integrated chip structure comprising a PID mitigation layer having a plurality of sub-layers.
- (16) FIGS. 43-49 illustrate some additional embodiments of a method of forming an integrated chip structure comprising a MIM structure having a PID mitigation layer.
- (17) FIG. 50 illustrates a flow diagram of some embodiments of a method of an integrated chip structure comprising a PID mitigation layer.

DETAILED DESCRIPTION

(18) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(19) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(20) The semiconductor industry has continually improved the performance of integrated chips by scaling the minimum feature size of integrated chip (IC) components according to Moore's law. As minimum feature sizes have decreased, the size of conductive interconnects on the back-end-of-the line (BEOL) and the space between the conductive interconnects has also decreased. The smaller conductive interconnects and space therebetween has increased a density of the interconnects, thereby improving a performance of a corresponding integrated chip.

(21) The high density of interconnects within modern-day integrated chips are made possible in-part by plasma processes (e.g., plasma etching and deposition processes). For example, plasma etching processes allow for trenches and/or via holes used in damascene processes to be formed with higher aspect ratios than non-plasma processes. Similarly, plasma deposition processes provide for improved gap fill over non-plasma processes. However, when plasma processes are used to form interconnect structures, charges from a plasma may flow through conductive interconnects to underlying semiconductor devices. The charges can stress and/or damage gate dielectrics in the underlying semiconductor devices, thereby degrading a quality of the gate dielectrics and resulting in a high gate leakage current, device failure, and/or other reliability issues. For core NMOS/PMOS devices, which may have a current leakage failure rate that are less than 5%, such stress and damage can make it difficult for the devices to pass reliability qualification.

(22) The present disclosure relates to a method of forming an interconnect structure having a plasma induced damage (PID) mitigation layer configured to reduce plasma induced damage. In some embodiments, the method forms a lower interconnect within a lower inter-level dielectric (ILD) structure formed over a substrate. A plasma induced damage (PID) mitigation layer is formed over the lower interconnect and the lower ILD structure using a plasma deposition process performed at a relatively high pressure. The relatively high pressure increases collisions between charged particles within a plasma, thereby reducing an energy of charged particles within the plasma and mitigating plasma induced damage caused during formation of the PID mitigation layer. The PID mitigation layer is able to prevent plasma damage during the subsequent formation of overlying layers (e.g., an overlying metal nitride layer, an upper interconnect, etc.), thereby mitigating plasma induced damage on the integrated chip structure.

(23) FIG. 1 illustrates a cross-sectional view of some embodiments of an integrated chip structure 100 comprising a plasma induced damage (PID) mitigation layer configured to reduce plasma

induced damage.

(24) The integrated chip structure **100** includes a lower inter-level dielectric (ILD) structure **104L** disposed over a substrate **102**. In some embodiments, the lower ILD structure **104L** comprises one or more lower ILD layers stacked onto one another. The lower ILD structure **104L** surrounds one or more lower interconnects **106**. In some embodiments, the one or more lower interconnects **106** may comprise a first lower interconnect **108**, a second lower interconnect **110**, and a third lower interconnect **112**. In some embodiments, the first lower interconnect **108** may comprise or be a conductive contact or a middle-end-of-the-line (MEOL) interconnect, the second lower interconnect **110** may comprise or be an interconnect wire, and the third lower interconnect **112** may comprise or be an interconnect via.

(25) In some embodiments, the one or more lower interconnects **106** are coupled to a semiconductor device **114** disposed on and/or within the substrate **102**. In some embodiments, the semiconductor device **114** may comprise a transistor device (e.g., a planar field effect transistor (FET), a FinFET, a gate all around (GAA) device, a nanosheet device, or the like). In other embodiments, the semiconductor device **114** may comprise a memory device (e.g., a ferroelectric field effect transistor (FeFET), a nanoparticle organic memory FET (NOMFET), or the like).

(26) A plasma induced damage (PID) mitigation layer **116** is arranged over the lower ILD structure **104L**. In some embodiments, the PID mitigation layer **116** comprises a conductive material and/or a metal. In some embodiments, the PID mitigation layer **116** may comprise a metal nitride, such as titanium nitride, tantalum nitride, or the like. In some embodiments, the PID mitigation layer **116** may comprise a metal nitride having a metal to nitrogen ratio that is greater than 1, that is between approximately 1 and approximately 1.5, that is between approximately 1 and approximately 1.1, or other similar values.

(27) In some embodiments, a metal nitride layer **118** is disposed over the PID mitigation layer **116**. One or more upper interconnects **119** are disposed within an upper ILD structure **104U** disposed over the metal nitride layer **118**. The one or more upper interconnects **119** may comprise a first upper interconnect **120** arranged on the one or more lower interconnects **106** and a second upper interconnect **122** disposed on the first upper interconnect **120**. In some embodiments, the first upper interconnect **120** extends from between sidewalls of the PID mitigation layer **116** and/or the metal nitride layer **118** to above the PID mitigation layer **116**.

(28) In some embodiments, the PID mitigation layer **116** may be formed by a first plasma deposition process performed at a first pressure that is relatively high (e.g., greater than or equal to approximately 15 mTorr). The relatively high first pressure increases ion collisions within a first plasma of the first plasma deposition process, thereby reducing an energy of the ions and mitigating plasma induced damage caused during formation of the PID mitigation layer **116**. The first pressure also causes the PID mitigation layer **116** to have a porous structure that gives the PID mitigation layer **116** a first density. In some embodiments, the porous structure includes a plurality of columnar structures.

(29) In some embodiments, the metal nitride layer **118** may be formed using a second plasma deposition process performed at a second pressure that is less than the first pressure. The second pressure causes the metal nitride layer **118** to have a lower roughness than the PID mitigation layer **116** and a second density that is greater than the first density. The lower roughness of the metal nitride layer **118** improves a planarity of overlying layers, thereby improving a process window of the overlying layers.

(30) The PID mitigation layer **116** is configured to prevent charged particles (e.g., ions) within a second plasma of the second plasma deposition process from flowing within the one or more lower interconnects **106**. For example, in some embodiments the PID mitigation layer **116** may comprise a conductive layer that has a lower resistance than the one or more lower interconnects **106**. Because the PID mitigation layer **116** has a lower resistance than the one or more lower interconnects **106**, charged particles within the second plasma will flow across the PID mitigation

layer **116** rather than through the one or more lower interconnects **106**. By preventing charged particles within the second plasma from flowing within the one or more lower interconnects **106**, the PID mitigation layer **116** reduces plasma induced damage on the semiconductor device **114**.

(31) FIG. 2 illustrates a cross-sectional view of some additional embodiments of an integrated chip structure **200** comprising a PID mitigation layer.

(32) The integrated chip structure **200** includes a lower ILD structure **104L** disposed over a substrate **102**. The lower ILD structure **104L** comprises a plurality of lower dielectric layers **104a-104b** stacked onto one another. In some embodiments, the plurality of lower dielectric layers **104a-104b** may comprise one or more of silicon dioxide, SiCOH, borophosphate silicate glass (BSG), phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), fluorosilicate glass (FSG), undoped silicate glass (USG), or the like. In some embodiments, the lower ILD structure **104L** may further comprise a contact etch stop layer (CESL) **202** disposed on the substrate **102**. In some embodiments, the CESL **202** may comprise one or more of a nitride (e.g., silicon nitride, silicon oxynitride, etc.), a carbide (e.g., silicon carbide, silicon oxycarbide, etc.), or the like.

(33) The lower ILD structure **104L** surrounds one or more lower interconnects **106**. In some embodiments, the one or more lower interconnects **106** may comprise a first lower interconnect **108**, a second lower interconnect **110**, and a third lower interconnect **112**. In some embodiments, the one or more lower interconnects **106** may comprise a conductive material, such as tungsten, copper, ruthenium, tantalum, titanium, or the like. In some embodiments, one or more of the one or more lower interconnects **106** may comprise a barrier layer surrounding a conductive core. For example, the third lower interconnect **112** may comprise a barrier layer **112b** surrounding a conductive core **112c**. In some embodiments, the barrier layer **112b** may comprise a metal nitride (e.g., titanium nitride, tantalum nitride, or the like), while the conductive core **112c** may comprise a metal (e.g., copper, tungsten, aluminum, or the like).

(34) In some embodiments, the one or more lower interconnects **106** are coupled to a semiconductor device **114** disposed on and/or within the substrate **102**. In some embodiments, the semiconductor device **114** may comprise a gate electrode **204** disposed laterally between source/drain regions **206**. In various embodiments, the source/drain regions **206** may comprise a source or a drain, individually or collectively depending upon the context. The gate electrode **204** may be vertically separated from the substrate **102** by a gate dielectric **208**. In various embodiments, the gate electrode **204** may comprise polysilicon, a metal, or the like. In some embodiments, the gate dielectric **208** may comprise an oxide (e.g., silicon oxide), a high-k dielectric material (e.g., hafnium oxide), or the like.

(35) A PID mitigation layer **116** is disposed over the lower ILD structure **104L** and a metal nitride layer **118** is arranged over the PID mitigation layer **116**. In some embodiments, the PID mitigation layer **116** has a substantially flat upper surface extending between outermost sidewalls of the PID mitigation layer **116**. In some embodiments, the metal nitride layer **118** has a substantially flat upper surface extending between outermost sidewalls of the metal nitride layer **118**. In some embodiments, the PID mitigation layer **116** may have a first nitrogen content (e.g., in a range of between approximately 40% and approximately 50%) and the metal nitride layer **118** may have a second nitrogen content that is higher than the first nitrogen content. In some embodiments, the metal nitride layer **118** may have a lower dielectric constant than the PID mitigation layer **116**.

(36) In some embodiments, the PID mitigation layer **116** may have a first thickness **210** and the metal nitride layer **118** may have a second thickness **212**. In some embodiments, a ratio of the first thickness **210** to the second thickness **212** may be in a range of between approximately 3/25 and approximately 9/15. In some embodiments, the first thickness **210** may be in a range of between approximately 25 Angstroms (Å) and approximately 100 Å, between approximately 30 Å and approximately 90 Å, greater than approximately 30 Å, or other similar values. A thickness that is less than approximately 25 Å or 30 Å may be insufficient to prevent plasma induced damage. In some embodiments, the second thickness **212** may be in a range of between approximately 100 Å

and approximately 300 Å, between approximately 150 Å and approximately 250 Å, or other similar values.

(37) A first upper interconnect **120** extends from between sidewalls of the PID mitigation layer **116** to over a top of the PID mitigation layer **116**. In some embodiments, the PID mitigation layer **116** is arranged directly below peripheral regions of the first upper interconnect **120** and not directly below a central region of the first upper interconnect **120**, so that the PID mitigation layer **116** is completely laterally outside of the central region of the first upper interconnect **120**. A second upper interconnect **122** is arranged on the first upper interconnect **120**. In some embodiments, the second upper interconnect **122** may comprise a conductive core **122c** and a barrier layer **122b**. In some embodiments (not shown), the first upper interconnect **120** may also comprise a barrier layer extending along sidewalls of the first upper interconnect **120** that are over the PID mitigation layer **116**.

(38) An upper ILD structure **104U** laterally surrounds the first upper interconnect **120** and the second upper interconnect **122**. The upper ILD structure **104U** comprises one or more upper ILD layers. In some embodiments, the one or more upper ILD layers may comprise one or more of silicon dioxide, SiCOH, BSG, PSG, BPSG, FSG, USG, or the like. In some embodiments, the first upper interconnect **120** has an upper surface that is arranged directly between a lower surface of the upper ILD structure **104U** and the top surface of the PID mitigation layer **116** and/or the metal nitride layer **118**.

(39) FIG. 3 illustrates a cross-sectional view of some additional embodiments of an integrated chip structure **300** comprising a plurality of PID mitigation layers.

(40) The integrated chip structure **300** comprises a semiconductor device **114** disposed on and/or within a substrate **102**. In some embodiments, the semiconductor device **114** may comprise a field effect transistor (FET). In some embodiments, the semiconductor device **114** may comprise a ferroelectric field effect transistor (FeFET) device. In such embodiments, the semiconductor device **114** comprises a gate electrode **204** disposed between source/drain regions **206**. The gate electrode **204** is separated from the substrate **102** by a gate dielectric **208**. The gate electrode **204** is further separated from the gate dielectric **208** by a ferroelectric layer **302**. In some embodiments, the ferroelectric layer **302** may extend along a lower surface and along sidewalls of the gate electrode **204**. In various embodiments, the ferroelectric layer **302** may be or comprise hafnium oxide, hafnium-zirconium-oxide (e.g., $\text{Hf.sub.xZr.sub.1-xO.sub.y}$, $\text{Hf.sub.0.5Zr.sub.0.5O.sub.2}$, etc.), aluminum nitride doped with scandium, hafnium-zirconium-oxide doped with one or more dopants (e.g., aluminum, silicon, lanthanum, scandium, calcium, barium, gadolinium, yttrium, another suitable dopant, or any combination of the foregoing), beryllium oxide, zinc oxide, calcium oxide, strontium oxide, boron oxide, zirconium dioxide, or the like. In some embodiments, sidewall spacers **304** may be arranged along opposing sides of the ferroelectric layer **302**.

(41) A plurality of lower interconnects **106** are arranged within a lower ILD structure **104L** comprising a plurality of stacked lower ILD layers **104a-104c**. The plurality of lower interconnects **106** are coupled to the semiconductor device **114**. A PID mitigation layer **116** is arranged over the lower ILD structure **104L** and a metal nitride layer **118** is arranged over the PID mitigation layer **116**. A first upper interconnect **120** extends through the PID mitigation layer **116** and the metal nitride layer **118** to over the metal nitride layer **118**. A second upper interconnect **122** is arranged on the first upper interconnect **120**. An upper ILD structure **104U** is arranged around the first upper interconnect **120** and the second upper interconnect **122**.

(42) In some embodiments, the PID mitigation layer **116** and the metal nitride layer **118** may comprise a plurality of discrete segments that are laterally separated from one another by the upper ILD structure **104U**. By separating the discrete segments of the PID mitigation layer **116** and the metal nitride layer **118** from one another, the plurality of upper interconnects **119** are electrically isolated from one another. In some embodiments, the PID mitigation layer **116** and the metal nitride layer **118** may comprise segments that are completely confined below the first upper interconnect

120. In some embodiments, the PID mitigation layer **116** and the metal nitride layer **118** may comprise segments that laterally extend past an outermost sidewall of the first upper interconnect **120**. In some embodiments, the PID mitigation layer **116** and the metal nitride layer **118** may comprise segments that continuously extend between two adjacent ones of the plurality of upper interconnects **119**. In such embodiments, the two adjacent ones of the plurality of upper interconnects **119** are electrically coupled together.

(43) An additional PID mitigation layer **306** is arranged over the upper ILD structure **104U** and an additional metal nitride layer **308** is arranged over the additional PID mitigation layer **306**. A third upper interconnect **310** extends through the additional PID mitigation layer **306** and the additional metal nitride layer **308** to over the additional metal nitride layer **308**. An additional upper ILD structure **104U.sub.A** is arranged around the third upper interconnect **310**.

(44) FIG. **4** illustrates a cross-sectional view of some additional embodiments of an integrated chip structure **400** comprising a PID mitigation layer.

(45) The integrated chip structure **400** includes a lower ILD structure **104L** disposed over a substrate **102**. The lower ILD structure **104L** surrounds one or more lower interconnects **106** that are coupled to a semiconductor device **114** (e.g., a field effect transistor) disposed on or within the substrate **102**. A dielectric layer **402** is arranged over the lower ILD structure **104L**. The dielectric layer **402** may comprise an oxide (e.g., silicon oxide), a nitride (e.g., silicon nitride, silicon oxynitride, etc.), a carbide (e.g., silicon carbide, silicon oxycarbide, etc.), or the like.

(46) The dielectric layer **402** has a thickness **404**. In some embodiments, a ratio of the thickness **404** to a thickness of the PID mitigation layer **116** is in a range of between approximately $\frac{1}{2}$ and approximately 7. In some embodiments, the dielectric layer **402** may have a thickness **404** that is in a range of between approximately 50 Angstroms (Å) and approximately 200 Å, between approximately 100 Å and approximately 150 Å, or other similar values. A PID mitigation layer **116** is arranged over the dielectric layer **402** and a metal nitride layer **118** is disposed over the PID mitigation layer **116**.

(47) The dielectric layer **402** may be formed using a plasma deposition process that has a relatively low power. For example, in some embodiments the dielectric layer **402** may be formed at a power that is less than or equal to a power used to form the PID mitigation layer **116**. The low power of the plasma deposition process allow for the dielectric layer **402** to be formed without causing plasma induced damage on the semiconductor device **114**.

(48) One or more upper interconnects **119** are disposed within an upper ILD structure **104U** disposed over the metal nitride layer **118**. The one or more upper interconnects **119** comprise a first upper interconnect **120** and a second upper interconnect **122**. The first upper interconnect **120** extends through the dielectric layer **402**, the PID mitigation layer **116**, and the metal nitride layer **118** to contact the one or more lower interconnects **106**. In some embodiments, the dielectric layer **402** may be configured to act as an etch stop layer used during formation of the first upper interconnect **120**.

(49) FIG. **5A** illustrates a cross-sectional view of some additional embodiments of an integrated chip structure **500** comprising a PID mitigation layer.

(50) The integrated chip structure **500** includes a lower ILD structure **104L** disposed over a substrate **102**. The lower ILD structure **104L** surrounds one or more lower interconnects **106** that are coupled to a semiconductor device **114** disposed on or within the substrate **102**.

(51) A PID mitigation layer **116** is arranged over the lower ILD structure **104L**. A first upper interconnect **120** extends through the PID mitigation layer **116**. In some embodiments, the first upper interconnect **120** may extend from a lower surface contacting the one or more lower interconnects **106**, to directly between sidewalls of the PID mitigation layer **116**, and to directly over the PID mitigation layer **116**. In some embodiments, the first upper interconnect **120** contacts the sidewalls and the upper surface of the PID mitigation layer **116**. In some embodiments, the PID mitigation layer **116** may have a thickness **502** that is in a range of between approximately 100 Å

and approximately 500 Å, between approximately 200 Å and approximately 400 Å, or other similar values.

(52) Having the first upper interconnect **120** contact sidewalls and the upper surface of the PID mitigation layer **116** allows for the integrated chip structure **500** to be formed using a relatively simple fabrication process (e.g., excluding a further deposition process to form a metal nitride layer) that can reduce fabrication costs relative to the structure shown in FIG. **1**, for example. However, because the PID mitigation layer **116** is formed by a plasma deposition process performed at a high pressure, the PID mitigation layer **116** will reduce plasma induced damage on the semiconductor device **114**.

(53) FIG. **5B** illustrates a cross-sectional view of some additional embodiments of an integrated chip structure **504** comprising a PID mitigation layer.

(54) The integrated chip structure **504** includes a dielectric layer **402** disposed between the PID mitigation layer **116** and the lower ILD structure **104L**. The first upper interconnect **120** extends through the PID mitigation layer **116** and the dielectric layer **402**.

(55) FIG. **6** illustrates a cross-sectional view of some additional embodiments of an integrated chip structure **600** comprising a PID mitigation layer.

(56) The integrated chip structure **600** includes a lower ILD structure **104L** disposed over a substrate **102**. The lower ILD structure **104L** surrounds one or more lower interconnects **106** that are coupled to a semiconductor device **114** disposed on or within the substrate **102**.

(57) A PID mitigation layer **116** is arranged over the lower ILD structure **104L**. In some embodiments, a lower surface of the PID mitigation layer **116** may have a first concentration of nitrogen, while an upper surface of the PID mitigation layer **116** may have a second concentration of nitrogen that is different than (e.g., larger than) the first concentration. In some embodiments, the PID mitigation layer **116** may have a gradient concentration that continuously increases from the lower surface to the upper surface. In other embodiments, the PID mitigation layer **116** may comprise a plurality of different PID mitigation sub-layers **116a-116b** respectively having different nitrogen concentrations. In such embodiments, the overall concentration of the PID mitigation layer **116** may increase in a step-wise manner between the different nitrogen concentrations of the different PID mitigation sub-layers **116a-116b**.

(58) In some embodiments, the plurality of different PID mitigation sub-layers **116a-116b** may comprise a first PID mitigation sub-layer **116a** and a second PID mitigation sub-layer **116b**. The first PID mitigation sub-layer **116a** has a first nitrogen concentration and the second PID mitigation sub-layer **116b** has a second nitrogen concentration that is greater than the first nitrogen concentration. For example, the first nitrogen concentration may be in a range of between approximately 40% and approximately 45%, while the second nitrogen concentration may be in a range of between approximately 45% and approximately 50%.

(59) In some embodiments, a metal nitride layer **118** is arranged over the PID mitigation layer **116**. In some embodiments, the metal nitride layer **118** may have a higher nitrogen concentration than a maximum nitrogen concentration of the PID mitigation layer **116**. A first upper interconnect **120** is arranged on the metal nitride layer **118** and extends from over the metal nitride layer **118** to the one or more lower interconnects **106**.

(60) FIG. **7** illustrates a cross-sectional view of some additional embodiments of an integrated chip structure **700** comprising a PID mitigation layer.

(61) The integrated chip structure **700** includes a lower ILD structure **104L** disposed over a substrate **102**. The lower ILD structure **104L** surrounds one or more lower interconnects **106** that are coupled to a semiconductor device **114** disposed on or within the substrate **102**. The one or more lower interconnects **106** comprise a first lower interconnect **108**, a second lower interconnect **110**, and a third lower interconnect **112**. In some embodiments, a lower etch stop layer **702** may be arranged along a top of the second lower interconnect **110**. The third lower interconnect **112** may extend through the lower etch stop layer **702** to contact the second lower interconnect **110**. In some

embodiments, the lower etch stop layer **702** may comprise one or more of a nitride (e.g., silicon nitride, silicon oxynitride, etc.), a carbide (e.g., silicon carbide, silicon oxycarbide, etc.), or the like. (62) A dielectric layer **402** is arranged over the lower ILD structure **104L**, a PID mitigation layer **116** is arranged over the dielectric layer **402**, and a metal nitride layer **118** is arranged over the PID mitigation layer **116**. A first upper interconnect **120** extends through the dielectric layer **402**, the PID mitigation layer **116**, and the metal nitride layer **118** to contact the one or more lower interconnects **106**. A second upper interconnect **122** is arranged on the first upper interconnect **120**. In some embodiments, an upper etch stop layer **704** may be arranged along an upper surface of the first upper interconnect **120**. The second upper interconnect **122** extends through the upper etch stop layer **704** to contact the first upper interconnect **120**. In some embodiments, the upper etch stop layer **704** may comprise one or more of a nitride (e.g., silicon nitride, silicon oxynitride, etc.), a carbide (e.g., silicon carbide, silicon oxycarbide, etc.), or the like.

(63) In some embodiments, the first upper interconnect **120** may have lower sidewalls that are arranged between sidewalls of the PID mitigation layer **116** and upper sidewalls that are disposed over the PID mitigation layer **116**. In some embodiments, the lower sidewalls that are angled at a first angle α with respect to a horizontal line that is parallel to a top of the substrate **102**. The first angle α of the lower sidewalls causes a lower portion of the first upper interconnect **120** to have a width that increases as a distance from the substrate **102** increases. In some embodiments, the upper sidewalls are angled at a second angle β with respect to a top surface of the metal nitride layer **118**. The second angle β of the upper sidewalls causes an upper portion of the first upper interconnect **120** to have a width that decreases as a distance from the substrate **102** increases. In some embodiments, the second upper interconnect **122** may have sidewalls that are angled at a third angle γ with respect to an upper surface of the first upper interconnect **122**. The third angle γ of the sidewalls causes the second upper interconnect **122** to have a width that decreases as a distance from the substrate **102** decreases.

(64) In some embodiments, the first upper interconnect **120** may have different widths directly between sidewalls of the dielectric layer **402**, sidewalls of the PID mitigation layer **116**, and sidewalls of the metal nitride layer **118**. For example, the first upper interconnect **120** may have a first width **706** directly between the sidewalls of the dielectric layer **402**, a second width **708** directly between the sidewalls of the PID mitigation layer **116**, and a third width **710** directly between the sidewalls of the metal nitride layer **118**.

(65) It will be appreciated that while the disclosed PID mitigation layer illustrated in FIGS. 1-7 reduces the flow of charged particles through exposed surfaces of the one or more lower interconnects, that plasma induced damage may also occur in integrated chip structures that have a dielectric layer (e.g., an etch stop layer) covering the one or more lower interconnects. For example, forming a metal onto a dielectric layer that is over a lower interconnect may form a MIM (metal-insulator-metal) structure that is capacitively coupled to the one or more lower interconnects (e.g., that has a capacitance that causes charged particles to form within the one or more lower interconnects). The capacitive coupling can stress a gate dielectric of an underlying semiconductor device, thereby leading to plasma induced damage. The extent of plasma induced damage in such MIM structures may increase as a size of metal above and/or below a dielectric layer increases (since capacitance is proportional to a size of a conductive structure).

(66) In some embodiments, the disclosed PID mitigation layer may also be used in MIM structures to prevent plasma induced damage in underlying semiconductor devices. For example, FIG. 8 illustrates a cross-sectional view of some additional embodiments of an integrated chip structure **800** comprising a PID mitigation layer configured to reduce plasma induced damage on a MIM structure.

(67) The integrated chip structure **800** includes a lower ILD structure **104L** disposed over a substrate **102**. The lower ILD structure **104L** surrounds one or more lower interconnects **106** that are coupled to a semiconductor device **114** disposed on or within the substrate **102**. A PID

mitigation layer **116** is arranged over the lower ILD structure **104L**. In some embodiments, a dielectric layer **402** may be arranged between the PID mitigation layer **116** and the lower ILD structure **104L**. In some embodiments, a metal nitride layer **118** may be arranged over the PID mitigation layer **116**. A first upper interconnect **120** is arranged entirely over a top of the PID mitigation layer **116** and/or a top of the metal nitride layer **118**, as viewed in the cross-sectional view. A second upper interconnect **122** may be arranged on the first upper interconnect **120**.

(68) The first upper interconnect **120** is separated from the one or more lower interconnects **106** by the dielectric layer **402**, the PID mitigation layer **116**, and/or the metal nitride layer **118**. Without the PID mitigation layer **116**, a plasma deposition process used to form the first upper interconnect **120** could capacitively couple to the one or more lower interconnects **106**, resulting in plasma induced damage on the semiconductor device **114**. However, the PID mitigation layer **116** mitigates capacitive coupling between the one or more lower interconnects **106** and the first upper interconnect **120**, thereby mitigating plasma induced damage on the semiconductor device **114**.

(69) It will be appreciated that different ones of the PID mitigation film schemes illustrated in FIGS. **1-8** may be integrated within a same integrated chip structure. Integrating the different PID mitigation film schemes allows for flexibility of design, costs, and/or reliability. For example, in some embodiments lower interconnects may utilize a metal nitride layer to improve planarity while overlying interconnects may not utilize a metal nitride layer to improve cost and throughput (e.g., since process tolerances may be less stringent at higher interconnects).

(70) FIG. **9** illustrates a cross-sectional view of some additional embodiments of an integrated chip structure comprising a plurality of PID mitigation layers.

(71) The integrated chip structure **900** comprises a PID mitigation layer **116** arranged over a lower ILD structure **104L** surrounding one or more lower interconnects **106** coupled to a semiconductor device **114** disposed on and/or within a substrate **102**. A metal nitride layer **118** is arranged on the PID mitigation layer **116**. A first upper interconnect **120** extends through the PID mitigation layer **116** and the metal nitride layer **118** to over the metal nitride layer **118**. A second upper interconnect **122** is arranged on the first upper interconnect **120**. An upper ILD structure **104U** is arranged around the first upper interconnect **120** and the second upper interconnect **122**.

(72) A dielectric layer **402** is arranged over the upper ILD structure **104U**, an additional PID mitigation layer **306** is arranged over the dielectric layer **402**, and an additional metal nitride layer **308** is arranged over the additional PID mitigation layer **306**. A third upper interconnect **310** extends through the dielectric layer **402**, the additional PID mitigation layer **306**, and the additional metal nitride layer **308** to over the additional metal nitride layer **308**. An additional upper ILD structure **104U.sub.A** is arranged around the third upper interconnect **310**. In some embodiments, the PID mitigation layer **116** and the additional PID mitigation layer **306** may comprise a same material (e.g., a same metal nitride). In other embodiments, the PID mitigation layer **116** and the additional PID mitigation layer **306** may comprise different materials (e.g., different metal nitrides).

(73) FIGS. **10A-10B** illustrate graphs, **1000** and **1006**, showing some embodiments of exemplary gate leakage currents for transistor devices on a semiconductor body.

(74) Metal interconnect routing builds an antenna structure over FEOL devices (e.g., transistor devices). The antenna structures allow for charges to go through the interconnects and towards the FEOL devices, thereby applying an unwanted gate voltage to a gate electrode of the FEOL devices. The unwanted gate voltage can degrade a quality of a gate dielectric, thereby resulting in a high gate leakage current due to PID effect. FIG. **10A** illustrates a graph **1000** showing some embodiments of exemplary gate leakage currents for NMOS devices on a semiconductor body (e.g., a die, a wafer, or the like).

(75) In graph **1000**, the x-axis illustrates a gate leakage current value. The y-axis illustrates a percent of devices that have different gate leakage currents. A gate leakage current **1002** for transistor devices of integrated chip structures not having a disclosed PID mitigation layer is

substantially constant for most devices on a semiconductor body. However, the gate leakage current **1002** increases for some transistor devices (e.g., for about 3% to about 10% of transistor devices on the semiconductor body, for about 3% to about 5% of transistor devices, for about 3% of transistor devices, or other similar values). In contrast, a gate leakage current **1004** for devices of integrated chip structures having a disclosed PID mitigation layer is substantially constant for substantially all transistor devices on a semiconductor body (e.g., for less than approximately 1% of transistor devices on the semiconductor body). Therefore, the disclosed PID mitigation layer decreases gate leakage currents for NMOS devices over a semiconductor body (e.g., for about 3% to about 10% of transistor devices, for about 3% to about 5% of transistor devices, for about 3% of transistor devices, or other similar values).

(76) FIG. **10B** illustrates a graph **1006** showing some embodiments of exemplary gate leakage currents for PMOS devices on a semiconductor body.

(77) In graph **1006**, the x-axis illustrates a gate leakage current value. The y-axis illustrates a percent of devices that have different gate leakage currents. A gate leakage current **1008** for devices of integrated chip structures not having a disclosed PID mitigation layer increases for some transistor devices (e.g., for about 40% to about 50% of transistor devices on a semiconductor body, for about 45% to about 50% of transistor devices, for about 47% of transistor devices, or other similar values). In contrast, a gate leakage current **1010** for devices of integrated chip structures having a disclosed PID mitigation layer is substantially constant for substantially all devices on a semiconductor body (e.g., for less than approximately 2% of transistor devices on the semiconductor body, for less than approximately 1% of transistor devices, or other similar values). Therefore, the disclosed PID mitigation layer decreases gate leakage currents for PMOS devices over a semiconductor body (e.g., for about 40% to about 50% of transistor devices on a semiconductor body, for about 45% to about 50% of transistor devices, for about 47% of transistor devices, or other similar values).

(78) FIGS. **11-20** illustrate cross-sectional views **1100-2000** of some embodiments of a method of forming an integrated chip structure comprising a disclosed PID mitigation layer. Although FIGS. **11-20**, **21-28**, **29-34**, **35-42**, and **43-50** are described in relation to methods, it will be appreciated that the structures disclosed in the methods are not limited to such methods, but instead may stand alone as structures independent of the methods.

(79) As shown in cross-sectional view **1100** of FIG. **11**, a substrate **102** is provided. In various embodiments, the substrate **102** may be any type of semiconductor body (e.g., silicon, SiGe, SOI, etc.), such as a semiconductor wafer and/or one or more die on a wafer, as well as any other type of semiconductor and/or epitaxial layers, associated therewith. In some embodiments, one or more semiconductor devices **114** are formed on and/or within the substrate **102**. In some embodiments, forming the one or more semiconductor devices **114** includes forming a gate dielectric film over the substrate **102**, forming a gate electrode film over the gate dielectric film, and selectively patterning the gate dielectric film and the gate electrode film according to an etching process to form a gate electrode **204** separated from the substrate **102** by a gate dielectric **208**. Source/drain region **206** can be subsequently implanted into the substrate **102** on opposing sides of the gate electrode **204**.

(80) A first lower interconnect **108** is formed over the substrate **102**. In some embodiments, the first lower interconnect **108** may be formed using a damascene process (e.g., a single damascene process or a dual damascene process). The damascene process is performed by forming a first lower ILD layer **104a** over the substrate **102**, etching the first lower ILD layer **104a** to form a hole and/or a trench, and filling the hole and/or trench with a conductive material. In some embodiments, the first lower ILD layer **104a** may be deposited by a deposition process (e.g., a physical vapor deposition (PVD) process, a chemical vapor deposition (CVD) process, a plasma enhanced CVD (PE-CVD) process, or the like) and the conductive material may be formed using a deposition process and/or a plating process (e.g., electroplating, electro-less plating, etc.). In various embodiments, the first lower interconnect **108** may comprise tungsten, copper, or

aluminum copper, or the like.

(81) A second lower interconnect **110** is formed over the first lower interconnect **108**. In some embodiments, the second lower interconnect **110** may be formed by depositing a conductive material over the substrate **102** and by subsequently etching the conductive material to form the second lower interconnect **110**. In other embodiments (not shown), the second lower interconnect **110** may be formed by way of a damascene process.

(82) As shown in cross-sectional view **1200** of FIG. **12**, a second lower ILD layer **104b** is formed on and/or over the second lower interconnect **110** to form a lower ILD structure **104L**. In some embodiments, the second lower ILD layer **104b** may extend from around the second lower interconnect **110** to over the second lower interconnect **110**. The second lower ILD layer **104b** may be formed by a deposition process (e.g., a PVD process, a CVD process, a PE-CVD process, an ALD process, or the like).

(83) The lower ILD structure **104L** is subsequently patterned to form a lower interconnect opening **1202** exposing an upper surface of the second lower interconnect **110**. In some embodiments, the lower ILD structure **104L** may be patterned by selectively exposing the lower ILD structure **104L** to a first etchant **1204** according to a first mask **1206**.

(84) As shown in cross-sectional view **1300** of FIG. **13**, a first diffusion barrier layer **1302** is formed over the lower ILD structure **104L** and within the lower interconnect opening **1202**. The first diffusion barrier layer **1302** may be formed by a deposition process (e.g., a PVD process, a CVD process, a PE-CVD process, an ALD process, or the like). A first conductive material **1304** is formed onto the first diffusion barrier layer **1302** and within the lower interconnect opening **1202**. The first conductive material **1304** may be formed by a deposition process (e.g., a PVD process, a CVD process, a PE-CVD process, or the like) and/or by a plating process (e.g., electroplating, electro-less plating, etc.).

(85) As shown in cross-sectional view **1400** of FIG. **14**, a planarization process is performed (along line **1402**) to remove parts of the first conductive material (e.g., **1304** of FIG. **13**) and the first diffusion barrier layer (e.g., **1302** of FIG. **13**) from over the lower ILD structure **104L**. By removing parts of the first conductive material and the first diffusion barrier layer, the planarization process forms a third lower interconnect **112** comprising a barrier layer **112b** and a conductive core **112c**. In some embodiments, the planarization process may comprise a chemical mechanical planarization (CMP) process, a wet etching process, a dry etching process, a mechanical grinding process, or the like.

(86) As shown in cross-sectional view **1500** of FIG. **15A**, a PID mitigation layer **116** is formed onto the lower ILD structure **104L**. The PID mitigation layer **116** comprises a metal film. In some embodiments, the PID mitigation layer **116** may comprise a metal nitride, such as titanium nitride, tantalum nitride, or the like. In some embodiments, the PID mitigation layer **116** may be formed by a first plasma deposition process (e.g., a plasma enhanced PVD process, a plasma enhanced CVD process, a plasma sputtering process, or the like) performed at a first pressure. In such embodiments, the first pressure increases a number of collisions between charged particles **1504** of a plasma **1502**. The collisions reduce an energy of the charged particles **1504**, thereby reducing a charging effect that the charged particles **1504** have on the one or more lower interconnects **106** and reducing plasma induced damage on the semiconductor device **114**. The first pressure also gives the PID mitigation layer **116** a first nitrogen content.

(87) In some embodiments, the PID mitigation layer **116** may be formed at a sputtering gas pressure (e.g., at an argon sputtering gas pressure) of greater than approximately 20 mTorr, greater than approximately 15 mTorr, or the like. In some embodiments, the PID mitigation layer **116** may be formed over a time of between approximately 100 seconds and approximately 750 seconds, between approximately 200 seconds and approximately 700 seconds, or other similar values. Forming the PID mitigation layer **116** over such times results in the PID mitigation layer **116** having a metal to nitrogen ratio that is greater than 1, that is between approximately 1 and

approximately 1.1, that is equal to approximately 1.1, or other similar values.

(88) For example, FIG. 15B illustrates a graph 1506 showing an exemplary content of a PID mitigation layer as a function of deposition time. As shown in graph 1506, in some embodiments the PID mitigation layer 116 may comprise a metal value (e.g., a titanium value) shown by line 1508 and a nitrogen value shown by line 1510. In some additional embodiments, the PID mitigation layer 116 may further comprise oxygen (shown by line 1512) and/or silicon (shown by line 1514). In some embodiments, box 1516 illustrates a range over which the PID mitigation layer 116 may be formed. In some embodiments, within the box 1516 the metal value shown by line 1508 may have a value of between approximately 45% and approximately 50%. In some embodiments, within the box 1516 the nitrogen value shown by line 1510 may be between approximately 45% and approximately 50%.

(89) In some embodiments, the first pressure causes the PID mitigation layer 116 to be formed as a porous structure having a low density. In some embodiments, the first pressure may cause the PID mitigation layer 116 to form as fibrous and/or columnar structures. For example, FIG. 15C illustrates a cross-sectional view 1518 of some embodiments of a PID mitigation layer 116 formed at the first pressure. As shown in cross-sectional view 1518, the PID mitigation layer 116 comprises a plurality of columnar structures 1520 laterally adjacent to one another. The plurality of columnar structures 1520 are separate and distinct from one another so as to give the PID mitigation layer 116 a porous structure. In some embodiments, the plurality of columnar structures 1520 also have different heights, giving a top of the PID mitigation layer 116 a first roughness.

(90) As shown in cross-sectional view 1600 of FIG. 16, a metal nitride layer 118 is formed over the PID mitigation layer 116. The metal nitride layer 118 may be formed using a second plasma deposition process. The second plasma deposition process utilizes a plasma 1602 having charged particles 1604 to form the metal nitride layer 118 on the PID mitigation layer 116. The PID mitigation layer 116 prevents charged particles 1604 from the plasma 1602 from transferring to the one or more lower interconnects 106. In some embodiments, the PID mitigation layer 116 may comprise a conductive layer (e.g., a conductive metal nitride layer) that has a lower resistance than the one or more lower interconnects 106. Because the PID mitigation layer 116 has a lower resistance than the one or more lower interconnects 106, the charged particles 1604 within the plasma 1602 will flow across the PID mitigation layer 116 rather than flowing through the one or more lower interconnects 106, thereby mitigating plasma induced damage.

(91) In some embodiments, the second plasma deposition process may be performed at a second pressure that is less than the first pressure. For example, the metal nitride layer 118 may be formed at a sputtering gas pressure of less than or equal to approximately 7 mTorr, less than or equal to approximately 5 mTorr, or the like. Forming the metal nitride layer 118 at the second pressure causes the metal nitride layer 118 to have an upper surface with a second roughness that is less than the first roughness, thereby giving an upper surface of the metal nitride layer 118 a greater smoothness than an upper surface of the PID mitigation layer 116. In some embodiments, the second pressure also causes the metal nitride layer 118 to be formed to have a second nitrogen concentration that is higher than the first nitrogen concentration of the PID mitigation layer 116. In some embodiments, the metal nitride layer 118 may also be formed to have a higher density than the PID mitigation layer 116.

(92) As shown in cross-sectional view 1700 of FIG. 17, the metal nitride layer 118 and the PID mitigation layer 116 are patterned according to a second patterning process to form a first upper interconnect opening 1702. The first upper interconnect opening 1702 extends through the metal nitride layer 118 and the PID mitigation layer 116 to expose an upper surface of the third lower interconnect 112. In some embodiments, the metal nitride layer 118 and the PID mitigation layer 116 may be patterned by selectively exposing the metal nitride layer 118 and the PID mitigation layer 116 to a second etchant 1704 according to a second mask 1706. In some embodiments, the second patterning process may be performed at a relatively low power (e.g., less than or equal to

100 W), so as to avoid plasma induced damage.

(93) As shown in cross-sectional view **1800** of FIG. **18**, a second conductive material **1802** is formed onto the metal nitride layer **118** and within the first upper interconnect opening **1702**. In various embodiments, the second conductive material **1802** may comprise aluminum, ruthenium, tungsten, copper, or the like. The second conductive material **1802** may be formed by a deposition process (e.g., a PVD process, a CVD process, a PE-CVD process, or the like), and/or by a plating process (e.g., electroplating, electro-less plating, etc.). In some embodiments, the second conductive material **1802** may be formed at a relatively low power (e.g., less than or equal to 100 W), so as to avoid plasma induced damage.

(94) As shown in cross-sectional view **1900** of FIG. **19**, the second conductive material (e.g., **1802** of FIG. **18**) is patterned according to a third patterning process to form a first upper interconnect **120**. The first upper interconnect **120** contacts the third lower interconnect **112**. In some embodiments, the second conductive material may be patterned according to a third patterning process that selectively exposes the second conductive material to a third etchant **1902** according to a third mask **1904**. In some embodiments, the third patterning process may be performed at a relatively low power (e.g., less than or equal to 100 W), so as to avoid plasma induced damage.

(95) As shown in cross-sectional view **2000** of FIG. **20**, an upper ILD structure **104U** is formed over and laterally surrounding the first upper interconnect **120**. The upper ILD structure **104U** may comprise one or more upper ILD layers. In some embodiments, the upper ILD structure **104U** may be formed by one or more deposition processes (e.g., a PVD process, a CVD process, a PE-CVD process, or the like). The upper ILD structure **104U** may comprise one or more of silicon dioxide, SiCOH, BSG, PSG, BPSG, FSG, USG, or the like.

(96) A second upper interconnect **122** is formed within the upper ILD structure **104U**. The second upper interconnect **122** extends through the upper ILD structure **104U** to contact the first upper interconnect **120**. In some embodiments, the second upper interconnect **122** may be formed using a damascene process (e.g., a single damascene process or a dual damascene process). The damascene process is performed by etching the upper ILD structure **104U** to form a second upper interconnect opening (e.g., via hole and/or the trench), and filling the second upper interconnect opening with a third conductive material. In some embodiments, the third conductive material (e.g., tungsten, copper, aluminum, or the like) may be formed using a deposition process and/or a plating process (e.g., electroplating, electro-less plating, etc.).

(97) FIGS. **21-28** illustrate some additional embodiments of a method of forming an integrated chip structure comprising a PID mitigation layer.

(98) As shown in cross-sectional view **2100** of FIG. **21**, a semiconductor device **114** is formed on and/or within the substrate **102**. One or more lower interconnects **106** are formed within a lower ILD structure **104L** formed over the substrate **102**. In some embodiments, the one or more lower interconnects **106** may comprise a first lower interconnect **108**, a second lower interconnect **110**, and a third lower interconnect **112**. In some embodiments, the semiconductor device **114** and the one or more lower interconnects **106** may be formed as described in the description associated with FIGS. **10-14**.

(99) As shown in cross-sectional view **2200** of FIG. **22**, a dielectric layer **402** is formed onto the lower ILD structure **104L**. The dielectric layer **402** is formed using a deposition process that is performed at a relatively low power (e.g., less than approximately 100 W, less than approximately 200 W, or other similar values). The low power prevents plasma induced damage from occurring during formation of the dielectric layer **402**.

(100) As shown in cross-sectional view **2300** of FIG. **23**, a PID mitigation layer **116** is formed onto the dielectric layer **402**. In some embodiments, the PID mitigation layer **116** is formed using a first plasma deposition process performed at a first pressure (e.g., greater than approximately 20 mTorr, greater than approximately 15 mTorr, or the like). In such embodiments, the first pressure increases a number of collisions between charged particles **2304** of a plasma **2302**. The collisions reduce an

energy of the charged particles **2304**, thereby reducing plasma induced damage. In some embodiments, the first pressure may cause the PID mitigation layer **116** to form as a porous structure having fibrous and/or columnar structures.

(101) As shown in cross-sectional view **2400** of FIG. **24**, a metal nitride layer **118** is formed over the PID mitigation layer **116**. In some embodiments, the metal nitride layer **118** may be formed using a second plasma deposition process performed at a second pressure that is less than the first pressure. The second plasma deposition process utilizes a plasma **2402** having charged particles **2404** to form the metal nitride layer **118** on the PID mitigation layer **116**. The PID mitigation layer **116** prevents charged particles **2404** from the plasma **2402** from transferring to the one or more lower interconnects **106**.

(102) As shown in cross-sectional view **2500** of FIG. **25**, the metal nitride layer **118**, the PID mitigation layer **116**, and the dielectric layer **402** are patterned to form a first upper interconnect opening **2502**. The first upper interconnect opening **2502** extends through the metal nitride layer **118**, the PID mitigation layer **116**, and the dielectric layer **402** to expose an upper surface of the third lower interconnect **112**. In some embodiments, the metal nitride layer **118**, the PID mitigation layer **116**, and the dielectric layer **402** may be patterned by a first patterning process that selectively exposes the metal nitride layer **118**, the PID mitigation layer **116**, and the dielectric layer **402** to a first etchant **2504** according to a first mask **2506**.

(103) As shown in cross-sectional view **2600** of FIG. **26**, a second conductive material **2602** is formed onto the metal nitride layer **118** and within the first upper interconnect opening **2502**. In some embodiments, the second conductive material **2602** may be formed as described in the description associated with FIG. **18**.

(104) As shown in cross-sectional view **2700** of FIG. **27**, the second conductive material (e.g., **2602** of FIG. **26**) is patterned to form a first upper interconnect **120**. The first upper interconnect **120** extends through the first upper interconnect opening **2502** to contact the third lower interconnect **112**. In some embodiments, the second conductive material may be patterned by a second patterning process that selectively exposes the second conductive material to a second etchant **2702** according to a second mask **2704**.

(105) As shown in cross-sectional view **2800** of FIG. **28**, an upper ILD structure **104U** is formed over and laterally surrounding the first upper interconnect **120**. The upper ILD structure **104U** may be formed by a deposition process (e.g., a PVD process, a CVD process, a PE-CVD process, or the like). A second upper interconnect **122** is formed within the upper ILD structure **104U**. The second upper interconnect **122** extends through the upper ILD layer to contact the first upper interconnect **120**.

(106) FIGS. **29-34** illustrate some additional embodiments of a method of forming an integrated chip structure comprising a PID mitigation layer.

(107) As shown in cross-sectional view **2900** of FIG. **29**, a semiconductor device **114** is formed on and/or within the substrate **102**. One or more lower interconnects **106** are formed within a lower ILD structure **104L** formed over the substrate **102**. In some embodiments, the one or more lower interconnects **106** may comprise a first lower interconnect **108**, a second lower interconnect **110**, and a third lower interconnect **112**. In some embodiments, the semiconductor device **114** and the one or more lower interconnects **106** may be formed as described in the description associated with FIGS. **10-14**.

(108) As shown in cross-sectional view **3000** of FIG. **30**, a PID mitigation layer **116** is formed onto the lower ILD structure **104L**. In some embodiments, the PID mitigation layer **116** is formed using a first plasma deposition process performed at a first pressure (e.g., greater than approximately 20 mTorr, greater than approximately 15 mTorr, or the like). In such embodiments, the first pressure increases a number of collisions between charged particles **3004** of a plasma **3002**. The collisions reduce an energy of the charged particles **3004**, thereby reducing plasma induced damage. In some embodiments, the first pressure may cause the PID mitigation layer **116** to form as a porous

structure having fibrous and/or columnar structures.

(109) As shown in cross-sectional view **3100** of FIG. **31**, the PID mitigation layer **116** is patterned to form a first upper interconnect opening **3102**. The first upper interconnect opening **3102** extends through the PID mitigation layer **116** to expose an upper surface of the third lower interconnect **112**. In some embodiments, the PID mitigation layer **116** may be patterned by a first patterning process that selectively exposes the PID mitigation layer **116** to a first etchant **3104** according to a first mask **3106**. In some embodiments, the first patterning process may be performed at a relatively low power (e.g., less than or equal to 100 W), so as to avoid plasma induced damage on the semiconductor device **114**.

(110) As shown in cross-sectional view **3200** of FIG. **32**, a second conductive material **3202** is formed onto the PID mitigation layer **116** and within the first upper interconnect opening **3102**. In some embodiments, the second conductive material **3202** may be formed as described in the description associated with FIG. **18**.

(111) As shown in cross-sectional view **3300** of FIG. **33**, the second conductive material (e.g., **3202**) is patterned to form a first upper interconnect **120**. The first upper interconnect **120** extends through the first upper interconnect opening **3102** to contact the third lower interconnect **112**. In some embodiments, the second conductive material may be patterned by a second patterning process that selectively exposes the second conductive material to a second etchant **3302** according to a second mask **3304**.

(112) As shown in cross-sectional view **3400** of FIG. **34**, an upper ILD structure **104U** is formed over and laterally surrounding the first upper interconnect **120**. A second upper interconnect **122** is formed within the upper ILD structure **104U**. The second upper interconnect **122** extends through the upper ILD layer to contact the first upper interconnect **120**.

(113) FIGS. **35-42** illustrate some embodiments of a method of forming an integrated chip structure comprising a PID mitigation layer having a plurality of sub-layers.

(114) As shown in cross-sectional view **3500** of FIG. **35**, a semiconductor device **114** is formed on and/or within the substrate **102**. One or more lower interconnects **106** are formed within a lower ILD structure **104L** formed over the substrate **102**. In some embodiments, the one or more lower interconnects **106** may comprise a first lower interconnect **108**, a second lower interconnect **110**, and a third lower interconnect **112**. In some embodiments, the semiconductor device **114** and the one or more lower interconnects **106** may be formed as described in the description associated with FIGS. **10-14**.

(115) As shown in cross-sectional views **3600** of FIG. **36**, a first PID mitigation sub-layer **116a** is formed onto the lower ILD structure **104L**. In some embodiments, the first PID mitigation sub-layer **116a** may be formed to have a first nitrogen content. In some embodiments, the first PID mitigation sub-layer **116a** may be formed by a first plasma deposition process performed at a first pressure.

(116) As shown in cross-sectional views **3700** of FIG. **37**, a second PID mitigation sub-layer **116b** is formed onto the first PID mitigation sub-layer **116a** to form a PID mitigation layer **116**. In some embodiments, the second PID mitigation sub-layer **116b** may be formed to have a second nitrogen content that is larger than the first nitrogen content. In some embodiments, the second PID mitigation sub-layer **116b** may be formed by a second plasma deposition process performed at a second pressure that is smaller than the first pressure. In some embodiments, both the first pressure and the second pressure may be greater than approximately 20 mTorr, greater than approximately 15 mTorr, or the like.

(117) As shown in cross-sectional view **3800** of FIG. **38**, a metal nitride layer **118** is formed over the PID mitigation layer **116**. In some embodiments, the metal nitride layer **118** may be formed by a third plasma deposition process performed at a third pressure that is greater than the second pressure. The PID mitigation layer **116** prevents charged particles from a plasma used in the third plasma deposition process from transferring to the one or more lower interconnects **106**.

(118) As shown in cross-sectional view **3900** of FIG. **39**, the PID mitigation layer **116** and the metal nitride layer **118** are patterned to form a first upper interconnect opening **3902**. The first upper interconnect opening **3902** extends through the PID mitigation layer **116** and the metal nitride layer **118** to expose an upper surface of the third lower interconnect **112**. In some embodiments, the PID mitigation layer **116** and the metal nitride layer **118** may be patterned by a first patterning process that selectively exposes the PID mitigation layer **116** and the metal nitride layer **118** to a first etchant **3904** according to a first mask **3906**.

(119) As shown in cross-sectional view **4000** of FIG. **40**, a second conductive material **4002** is formed onto the metal nitride layer **118** and within the first upper interconnect opening **3902**. In some embodiments, the second conductive material **4002** may be formed as described in the description associated with FIG. **18**.

(120) As shown in cross-sectional view **4100** of FIG. **41**, the second conductive material (e.g., **4002** of FIG. **40**) is patterned to form a first upper interconnect **120**. The first upper interconnect **120** extends through the first upper interconnect opening **3902** to contact the third lower interconnect **112**. In some embodiments, the second conductive material may be patterned by a second patterning process that selectively exposes the second conductive material to a second etchant **4102** according to a second mask **4104**.

(121) As shown in cross-sectional view **4200** of FIG. **42**, an upper ILD structure **104U** is formed over and laterally surrounding the first upper interconnect **120**. A second upper interconnect **122** is formed within the upper ILD structure **104U**. The second upper interconnect **122** extends through the upper ILD layer to contact the first upper interconnect **120**.

(122) FIGS. **43-49** illustrate some additional embodiments of a method of forming an integrated chip structure comprising a MIM structure having a PID mitigation layer.

(123) As shown in cross-sectional view **4300** of FIG. **43**, a semiconductor device **114** is formed on and/or within the substrate **102**. One or more lower interconnects **106** are formed within a lower ILD structure **104L** formed over the substrate **102**. In some embodiments, the one or more lower interconnects **106** may comprise a first lower interconnect **108**, a second lower interconnect **110**, and a third lower interconnect **112**. In some embodiments, the semiconductor device **114** and the one or more lower interconnects **106** may be formed as described in the description associated with FIGS. **10-14**.

(124) As shown in cross-sectional view **4400** of FIG. **44**, a dielectric layer **402** is formed onto the lower ILD structure **104L**. The dielectric layer **402** is formed using a deposition process that is performed at a relatively low power (e.g., less than approximately 100 W, less than approximately 200 W, or other similar values). The low power prevents plasma induced damage from occurring during formation of the dielectric layer **402**.

(125) As shown in cross-sectional view **4500** of FIG. **45**, a PID mitigation layer **116** is formed onto the dielectric layer **402**. In some embodiments, the PID mitigation layer **116** is formed at using a first plasma deposition process performed at a first pressure (e.g., greater than approximately 20 mTorr, greater than approximately 15 mTorr, or the like). In some embodiments, the first pressure may cause the PID mitigation layer **116** to form as a porous structure having fibrous and/or columnar structures.

(126) As shown in cross-sectional view **4600** of FIG. **46**, a metal nitride layer **118** is formed over the PID mitigation layer **116**. In some embodiments, the metal nitride layer **118** may be formed using a second plasma deposition process performed at a second pressure that is less than the first pressure. The PID mitigation layer **116** prevents charged particles **2404** from a plasma used during the second plasma deposition process from transferring to the one or more lower interconnects **106**.

(127) As shown in cross-sectional view **4700** of FIG. **47**, a second conductive material **4702** is formed onto the metal nitride layer **118**. In some embodiments, the second conductive material **4702** may be formed as described in the description associated with FIG. **18**.

(128) As shown in cross-sectional view **4800** of FIG. **48**, the second conductive material (e.g., **4702**

of FIG. 47) is patterned to form a first upper interconnect **120**. The first upper interconnect **120** is over the metal nitride layer **118** as viewed in cross-sectional view **4800**. In some embodiments, the second conductive material may be patterned by using a first patterning process that selectively exposes the second conductive material to a first etchant **4802** according to a first mask **4804**.

(129) As shown in cross-sectional view **4900** of FIG. 49, an upper ILD structure **104U** is formed over and laterally surrounding the first upper interconnect **120**. A second upper interconnect **122** is formed within the upper ILD structure **104U**. The second upper interconnect **122** extends through the upper ILD layer to contact the first upper interconnect **120**.

(130) FIG. 50 illustrates a flow diagram of some embodiments of a method **5000** of forming an integrated chip structure comprising a disclosed PID mitigation layer.

(131) While method **5000** is illustrated and described herein as a series of acts or events, it will be appreciated that the illustrated ordering of such acts or events are not to be interpreted in a limiting sense. For example, some acts may occur in different orders and/or concurrently with other acts or events apart from those illustrated and/or described herein. In addition, not all illustrated acts may be required to implement one or more aspects or embodiments of the description herein. Further, one or more of the acts depicted herein may be carried out in one or more separate acts and/or phases.

(132) At act **5002**, one or more lower interconnects are formed within a lower ILD structure over a substrate. FIGS. 10-14 illustrate cross-sectional views **1000-1400** of some embodiments corresponding to act **5002**. FIGS. 21, 29, 35, and 43 illustrate cross-sectional views **2100**, **2900**, **3500**, and **4300** illustrate some alternative embodiments corresponding to act **5002**.

(133) At act **5004**, a dielectric layer may be formed over the one or more lower interconnects and the lower ILD structure in some embodiments. FIG. 22 illustrates a cross-sectional view **2200** of some embodiments corresponding to act **5004**. FIG. 44 illustrates a cross-sectional view **4400** of some alternative embodiments corresponding to act **5004**.

(134) At act **5006**, a PID mitigation layer is formed over the one or more lower interconnects, the lower ILD structure, and/or the dielectric layer at a first pressure. FIGS. 15A, 23, 30, 36-37, and 45 illustrate cross-sectional views **1500**, **2300**, **3000**, **3600-3700**, and **4500** of some embodiments corresponding to act **5006**.

(135) At act **5008**, a metal nitride layer is formed over the PID mitigation layer at a second pressure that is less than the first pressure, in some embodiments. FIGS. 16, 24, 38, and 46 illustrate cross-sectional views **1600**, **2400**, **3800**, and **4600** of some embodiments corresponding to act **5008**.

(136) At act **5010**, the metal nitride layer, the PID mitigation layer, and/or the dielectric layer are patterned to form an upper interconnect opening exposing the one or more lower interconnects, in some embodiments. FIGS. 17, 25, 31, and 39 illustrate cross-sectional views **1700**, **2500**, **3100**, and **3900** of some embodiments corresponding to act **5010**.

(137) At act **5012**, a first upper interconnect is formed within the upper interconnect opening and/or over the PID mitigation layer. FIGS. 18-19 illustrate cross-sectional views **1700-1800** of some embodiments corresponding to act **5012**. FIGS. 26-27 illustrate cross-sectional views **2600-2700** of some alternative embodiments corresponding to act **5012**. FIGS. 32-33 illustrate cross-sectional views **3200-3300** of some alternative embodiments corresponding to act **5012**. FIGS. 40-41 illustrate cross-sectional views **4000-4100** of some alternative embodiments corresponding to act **5012**. FIGS. 47-48 illustrate cross-sectional views **4700-4800** of some alternative embodiments corresponding to act **5012**.

(138) At act **5014**, a second upper interconnect is formed within an upper ILD structure that is formed over and laterally around the first upper interconnect. FIGS. 20, 28, 34, 42, and 49 illustrate cross-sectional views **2000**, **2800**, **3400**, **4200**, and **4900** of some embodiments corresponding to act **5014**.

(139) Accordingly, the present disclosure relates to an integrated chip structure comprising a plasma induced damage (PID) mitigation layer configured to reduce plasma induced damage.

(140) In some embodiments, the present disclosure relates to an integrated chip structure. The integrated chip structure includes a substrate; one or more lower interconnects disposed within a lower inter-level dielectric (ILD) structure over the substrate; a plasma induced damage (PID) mitigation layer disposed over the lower ILD structure, the PID mitigation layer having a porous structure including a metal; and a first upper interconnect laterally surrounded by an upper ILD structure over the PID mitigation layer, the first upper interconnect extending from over the PID mitigation layer to the one or more lower interconnects. In some embodiments, the integrated chip structure further includes a metal nitride layer vertically between the PID mitigation layer and the upper ILD structure, the first upper interconnect extending from within the upper ILD structure to through the PID mitigation layer and the metal nitride layer. In some embodiments, the PID mitigation layer has a lower concentration of nitrogen than the metal nitride layer. In some embodiments, the PID mitigation layer has a substantially flat upper surface extending between outermost sidewalls of the PID mitigation layer. In some embodiments, the PID mitigation layer includes titanium nitride or tantalum nitride. In some embodiments, the PID mitigation layer has a ratio of the metal to nitrogen that is in a range that is between approximately 1 and approximately 1.5. In some embodiments, the integrated chip structure further includes a dielectric layer arranged between the lower ILD structure and the PID mitigation layer, the first upper interconnect continuously extending from directly above the PID mitigation layer to physically contact the one or more lower interconnects. In some embodiments, the integrated chip structure further includes a dielectric layer arranged over the upper ILD structure; a second PID mitigation layer disposed over the dielectric layer, the second PID mitigation layer having a second porous structure having a second metal and nitrogen; and a third upper interconnect disposed within an additional upper ILD structure over the second PID mitigation layer, the third upper interconnect extending from within the additional upper ILD structure to through the second PID mitigation layer and the dielectric layer.

(141) In other embodiments, the present disclosure relates to an integrated chip structure. The integrated chip structure includes one or more lower interconnects disposed within a lower inter-level dielectric (ILD) structure over a substrate; a plasma induced damage (PID) mitigation layer disposed over the lower ILD structure, the PID mitigation layer including a metal nitride having a metal to nitrogen ratio that is greater than 1; and a first upper interconnect disposed within an upper ILD structure over the PID mitigation layer, the first upper interconnect extending through the upper ILD structure and the PID mitigation layer to contact the one or more lower interconnects. In some embodiments, the first upper interconnect continuously extends from directly between sidewalls of the PID mitigation layer to directly over a top surface of the PID mitigation layer. In some embodiments, the first upper interconnect has an upper surface that is directly between a lower surface of the upper ILD structure and the top surface of the PID mitigation layer. In some embodiments, the integrated chip structure further includes a metal nitride layer vertically between the PID mitigation layer and the upper ILD structure, wherein the metal nitride layer has a higher concentration of nitrogen than the PID mitigation layer. In some embodiments, the PID mitigation layer has a thickness of greater than approximately 30 Angstroms.

(142) In yet other embodiments, the present disclosure relates to a method of forming an integrated chip structure. The method includes forming one or more lower interconnects within a lower dielectric (ILD) structure over a substrate; forming a plasma induced damage (PID) mitigation layer over the lower ILD structure, the PID mitigation layer including a metal nitride formed at a first pressure; forming a metal nitride layer over the PID mitigation layer, wherein the metal nitride layer is formed at a second pressure that is less than the first pressure; patterning the PID mitigation layer and the metal nitride layer to form an upper interconnect opening; and forming a conductive material within the upper interconnect opening and over the metal nitride layer. In some embodiments, the PID mitigation layer includes a porous structure having fibrous or columnar structures. In some embodiments, the method further includes patterning the conductive material to

form a first upper interconnect; and forming an upper inter-level dielectric (ILD) structure over the first upper interconnect. In some embodiments, the first pressure is greater than approximately 20 mTorr. In some embodiments, the PID mitigation layer has a lower concentration of nitrogen than the metal nitride layer. In some embodiments, the PID mitigation layer has a nitrogen concentration of between approximately 40% and approximately 50%. In some embodiments, the PID mitigation layer and the metal nitride layer are formed using plasma deposition processes.

(143) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method of forming an integrated chip structure, comprising: forming one or more lower interconnects within a lower inter-level dielectric (ILD) structure over a substrate; forming a plasma induced damage (PID) mitigation layer over the lower ILD structure, wherein the PID mitigation layer comprises a metal nitride formed at a first pressure; forming a metal nitride layer over the PID mitigation layer, wherein the metal nitride layer is formed at a second pressure that is less than the first pressure, and wherein the PID mitigation layer has a different concentration of nitrogen than the metal nitride layer; patterning the PID mitigation layer and the metal nitride layer to form an upper interconnect opening; and forming a conductive material within the upper interconnect opening and over the metal nitride layer.
2. The method of claim 1, wherein the conductive material has a lower surface physically contacting a top of the metal nitride layer.
3. The method of claim 1, further comprising: patterning the conductive material to form a first upper interconnect; and forming an upper inter-level dielectric (ILD) structure over the first upper interconnect.
4. The method of claim 1, wherein the first pressure is greater than approximately 20 mTorr and the second pressure is less than or equal to approximately 7 mTorr.
5. The method of claim 1, wherein the PID mitigation layer has a lower concentration of nitrogen than the metal nitride layer.
6. The method of claim 1, wherein the PID mitigation layer comprises a metal nitride having a metal to nitrogen ratio that is greater than 1.
7. The method of claim 1, wherein the upper interconnect opening exposes an upper surface of the lower ILD structure.
8. A method of forming an integrated chip structure, comprising: forming one or more lower interconnects within a lower inter-level dielectric (ILD) structure over a substrate; forming a plasma induced damage (PID) mitigation layer over the lower ILD structure, wherein the PID mitigation layer comprises a metal nitride formed at a first pressure; forming a metal nitride layer over the PID mitigation layer, wherein the metal nitride layer is formed at a second pressure that is less than the first pressure, and wherein the PID mitigation layer has a different concentration of nitrogen than the metal nitride layer; patterning the PID mitigation layer and the metal nitride layer to form an opening; forming a conductive material within the opening and over the metal nitride layer; patterning the conductive material according to a masking layer to form an upper interconnect; and forming an upper ILD structure over and along sidewalls of the upper interconnect.

9. The method of claim 8, wherein the PID mitigation layer is formed to have a first roughness and the metal nitride layer is formed to have a second roughness that is less than the first roughness.
 10. The method of claim 8, wherein the PID mitigation layer comprises a metal nitride having a metal to nitrogen ratio that is greater than 1.
 11. The method of claim 8, wherein the upper interconnect has a first width directly between sidewalls of the PID mitigation layer and a second width measured along a lower surface contacting a top of the metal nitride layer, the second width being larger than the first width.
 12. The method of claim 8, wherein the metal nitride layer has a higher concentration of nitrogen than the PID mitigation layer.
 13. The method of claim 8, wherein the PID mitigation layer further comprises silicon and oxygen.
 14. The method of claim 8, wherein the opening exposes an upper surface of the lower ILD structure.
 15. A method of forming an integrated chip structure, comprising: forming one or more lower interconnects within a lower inter-level dielectric (ILD) structure over a substrate; forming a plasma induced damage (PID) mitigation layer over the lower ILD structure, wherein the PID mitigation layer is formed at a first pressure; forming a metal nitride layer over the PID mitigation layer, wherein the metal nitride layer is formed at a second pressure that is less than the first pressure, and wherein the PID mitigation layer has a different concentration of nitrogen than the metal nitride layer; patterning the PID mitigation layer and the metal nitride layer to form an upper interconnect opening; and forming a conductive material within the upper interconnect opening and over the metal nitride layer.
 16. The method of claim 15, wherein the upper interconnect has a lower surface contacting a top of the metal nitride layer.
 17. The method of claim 15, wherein the PID mitigation layer further comprises silicon.
 18. The method of claim 15, wherein the upper interconnect opening exposes an upper surface of the lower ILD structure that is laterally between the PID mitigation layer and an outer sidewall of the one or more lower interconnects.
 19. The method of claim 15, wherein the PID mitigation layer and the metal nitride layer are formed to have a different upper surface roughness.
 20. The method of claim 15, wherein the PID mitigation layer and the metal nitride layer are a same chemical compound and have the different concentration of nitrogen.
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